Sequence Alignments

Sequence alignments used to be a synonym to Bioinformatics

Some people still think that:

Bioinformatics == Alignment Algos

What are sequence alignments?

What is a sequence alignment

A way to arrange sequences to identify regions of similarity:

```
ATGCAAAC-AG
|||| .|. ||
ATGC-TATTAG
```

Similarity may be a consequence of functional, structural or evolutionary relationships

There are several different ways of displaying and representing the information of alignments.

Alignment applications

Compare sequences

1. Two sequences: **pairwise alignment**
2. More than two: **multiple sequence alignment**

Search for matches in big datasets:

1. Local similarities: **BLAST** (Basic Local Alignment Search Tool)
2. Match reads against a known genome: **short-read aligners**

How to pick the "correct" alignment?

Suppose that the sequence `ATGAA` can be aligned to the following alternatives:

```
ATGAA    ATGAA    ATGAA    AT-GAA
|.|.|    |||.|    |||.|    ||  |||
ACGCA    ATGCA    ATGTA    ATCGAA

  1        2        3        4
```

Which one do you think is "correct" and why?

How are alignments computed

Values are associated with:

1. an **exact match**: a positive score (5)
2. **a mismatch**: a negative score (penalty), may depend on what is mismatching (-4)
3. a **gap opening**: usually the most penalized action (-10)
4. a **gap extension**: making the gap longer (-0.5)

Adding up the values is called **scoring the alignment**.

The aligner find the arrangment of maximal score.

Aligners find the arrangement that produces the maximal score.

The scoring "drives" the alignment.

The truth about alignments

Alignments are misunderstood and misused. A lot.

It is **easy** to align similar sequences: scoring barely matters, very different scoring will produce the same alignments. *Consistent results:* 😄

It is **not so easy** to align dissimilar sequences: tiny changes in the scoring can produce wildly different alignments: *Why did you use that scoring?* 😓

What is the "right" alignment?

Any two sequences can be aligned.

The alignment score represents the sum of the each match/mismatch/gap/gap extension

Aligners find the arrangement that the produce the largest alignment score.

There is no such thing as **the best** alignment.

The **alignment** represents the **scoring matrix**.

Other considerations

Most aligners will only report alignments that make some sense and usually the longest alignment in a region

The above is not that simple as it sounds!

Alignment is a measure of similarity but not homology (shared ancestry).

What is a scoring matrix?

Different scoring matrices may produce different alignments.

Rows/columns represent the rewards and penalties.
`A` replaced by `A` gets `5` points.
`A` replaced by `T` gets `-4` points.

```
    A   T   G   C
A   5  -4  -4  -4
T  -4   5  -4  -4
G  -4  -4   5  -4
C  -4  -4  -4   5
```

ftp://ftp.ncbi.nih.gov/blast/matrices/

Who makes the scores and how?

Most scores are determined empirically from existing sequence comparisons.

They represent the probability of observing substitutions of a type for known sequences.

Protein alignments have many different scoring matrices to choose from. It matters a lot which scoring matrix you choose.

See **BLOSUM** vs **PAM** matrices.

For DNA alignments the scoring is usually simpler.

Score the alignments

Match=**5**, Mismatch=**-4**
Gap open=**-10**, Gap extend=**-0.5**

```
ATGAA    ATGAA    ATGAA    AT-GAA
|.|.|    |||.|    |||.|    ||  |||
ACGCA    ATGCA    ATGTA    ATCGAA

  ?        ?        ?        ?
```

Write the score under each alignment.

More on scoring

It is possible to pick meaningless scoring parameters that will lead to meaningless alignments.

In the majority use cases, we leave them on defaults or use a known matrix (BLOSUM, PAM)

The scoring matrix should be negative when summed by row/column otherwise may produce nonsense.

Gaps at the end may be treated differently than **gaps in the middle** (there is a biologically relevant rationale for this).

How are alignments displayed?

Graphical:

```
ATGC--ACAAG
||||  | .||
ATGCTTA-TAG
```

CIGAR (Compact Idiosyncratic Gapped Alignment Report) of the bottom relative to the top sequence:

4 matches, 2 insertions, 1 match, 1 deletion, 1 mismatch, 2 matches

```
4M2I1M1D1X2M
```

"Idiosyncratic" allright

```
ATGC--ACAAG
||||  | .||
ATGCTTA-TAG
```

Multiple CIGAR versions exist:

```
4M2I1M1D1X2M      # The GOOD
4M2IMDX2M         # The BAD (drops the 1)
4M2IMD3M          # And the UGLY (M match or mismatch)
```

No really, our founding fathers thought that using `M` to represent *match or mismatch* was a good idea. It is the *standard* though it is being replaced(slowly).

Alignment strategies

1. Global alignments

```
THISISALONGERSEQUENCEALIGNEDAGAINSTASHORTSEQUENCE
-------LONGER--------A---N-D-------ASHORT--------
```

2. Local alignments

```
LONGER
LONGER
```

3. Semi-global (global-local alignments)

```
LONGERSEQUENCEALIGNEDAGAINSTASHORT
LONGER--------A---N-D-------ASHORT
```

Alignment algorithms

1. Optimal. Mathematically precise and guarantee correctness.
2. Near-optimal algorithms. Much more efficient and almost always also correct.

Optimal alignments are usually computationally **very** demanding.

Most techniques rely on near-optimal aligners.

Helper aligners

We have two helper methods to allow you to run alignments at the command line. These use the aligners from the **EMBOSS** package.

```
# Store the program in the bin folder.
mkdir -p ~/bin

# Install the wrapper for the EMBOSS alignment tools.
curl http://data.biostarhandbook.com/align/global-align.sh
curl http://data.biostarhandbook.com/align/local-align.sh

# Make the scripts executable.
chmod +x ~/bin/*-align.sh
```

See the book chapter for the commands.

Global Alignments

Run:

```
global-align.sh THISLINE ISALIGNED
```

Produces:

```
a                    1 THISLI--NE-      8
                         ||.:  ||
b                    1 --ISALIGNED      9
```

with the scoring:

```
# Identity:        4/11 (36.4%)
# Similarity:      5/11 (45.5%)
# Gaps:            5/11 (45.5%)
```

Local Alignments

Run:

```
local-align.sh THISLINE ISALIGNED
```

Produces:

```
a                    7 NE       8
                       ||
b                    7 NE       8
```

with the scoring:

```
# Identity:        2/2 (100.0%)
# Similarity:      2/2 (100.0%)
# Gaps:            0/2 ( 0.0%)
```

Explore the parameters

Alignment may or may not change when you change the scoring. Why?

```
local-align.sh THISLINE ISALIGNED --gapopen 0
local-align.sh THISLINE ISALIGNED --gapopen 1
local-align.sh THISLINE ISALIGNED --gapopen 2
local-align.sh THISLINE ISALIGNED --gapopen 3
```

When do you recover the original alignment?

Download different substitution matrices

```
wget ftp://ftp.ncbi.nlm.nih.gov/blast/matrices/BLOSUM30
wget ftp://ftp.ncbi.nlm.nih.gov/blast/matrices/BLOSUM62
wget ftp://ftp.ncbi.nlm.nih.gov/blast/matrices/BLOSUM90
```

Look at each matrix:

```
cat BLOSUM30 | head
```

What do you get:

```
local-align.sh THISLINE ISALIGNED -data BLOSUM30
local-align.sh THISLINE ISALIGNED -data BLOSUM62
local-align.sh THISLINE ISALIGNED -data BLOSUM90
```

Limitations of alignments

Alignments are a mathematical concept.

They minimize a score under the assumtpion that simplest explanation is correct.

Alignments can be biologically incorrect - we always need additional evidence.

See the book chapter on Misleading Alignments.